Mark Zhou

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Education

University of Wisconsin–Madison — Madison, WI
B.S. Computer Sciences, College of Letters & Science — *Sep. 2026 – May 2030 (expected)*

George School — Newtown, PA
High School Diploma, College/University Preparatory Program — *Sep. 2021 – May 2026*

Research

Research Collaborator, Gerych Lab — Worcester Polytechnic Institute
Representation Learning & Multimodal ML (Remote); Advisor: Prof. Walter Gerych — *Apr. 2026 – Present*

- Leading implementation and experimental design for ConceptBasis (demo), a paper on VLM concept directions
- Built the end-to-end research stack: Gemma-4 labeling, SigLIP2 adapters, and sealed class-disjoint evaluation
- Combined reverse-ridge regression with an orthogonality loss, raising compositional Recall@5 from 19.8% to 74.5%
- Designed five-seed, class-disjoint evaluation across concept prediction, direction overlap, and compositional retrieval

Projects

RotateMate – Image Orientation Classifier — github.com/markjouh/RotateMate-model
Personal Project — *2025 – Present*

- Built an end-to-end iOS ML system with custom data, MobileNetV4 training, and SwiftUI deployment
- Quantized a 3M-parameter classifier to 8-bit Core ML and optimized preprocessing for <1 ms on-device inference

Tiaoqi Master – Chinese Checkers iOS App — github.com/markjouh/tiaoqi-master
Swift Student Challenge 2026 Winner — *2025 – 2026*

- Built a native Swift 6/SwiftUI learning app with a play-as-you-learn tutorial and adaptive on-device AI
- Implemented time-budgeted iterative-deepening minimax with alpha-beta pruning and heuristic move ordering
- Engineered contextual move commentary and procedural audio synthesis with AVFoundation

Competitive Programming Library — github.com/markjouh/cp-library
Open Source Project — *2023 – Present*

- Built a CI-verified C++ library spanning data structures, graphs/flow, strings, number theory, and heuristics
- Automated regression tests against Library Checker and Aizu Online Judge through GitHub Actions

Technical Skills

ML Systems: GPU inference profiling, CUDA, vLLM, Core ML, PyTorch, experiment orchestration
ML Methods: Vision-language models, contrastive learning, adapter training, multimodal labeling and evaluation
Languages: C++, Python, Swift, C, Java, JavaScript, HTML/CSS
Frameworks & Tools: AVFoundation, SwiftUI, GitHub Actions, Linux, Git, Xcode

Honors & Leadership

Competitive Programming: USACO Platinum; Codeforces Master (2219, top 1%)
Product Awards: Apple Swift Student Challenge Winner (2026); c0mpiled Startup School Hackathon II (YC Startup School) – 2nd Place & Most Venture-Backable ($1,500)
Contest Wins: 1st Place – Carnegie Mellon CMIMC Programming (2x), UPenn PClassic Advanced, Massachusetts Computer Science Olympiad, and TJIOI
Leadership: Algorithms Club President & Co-Founder; Mathematics Tutor, George School